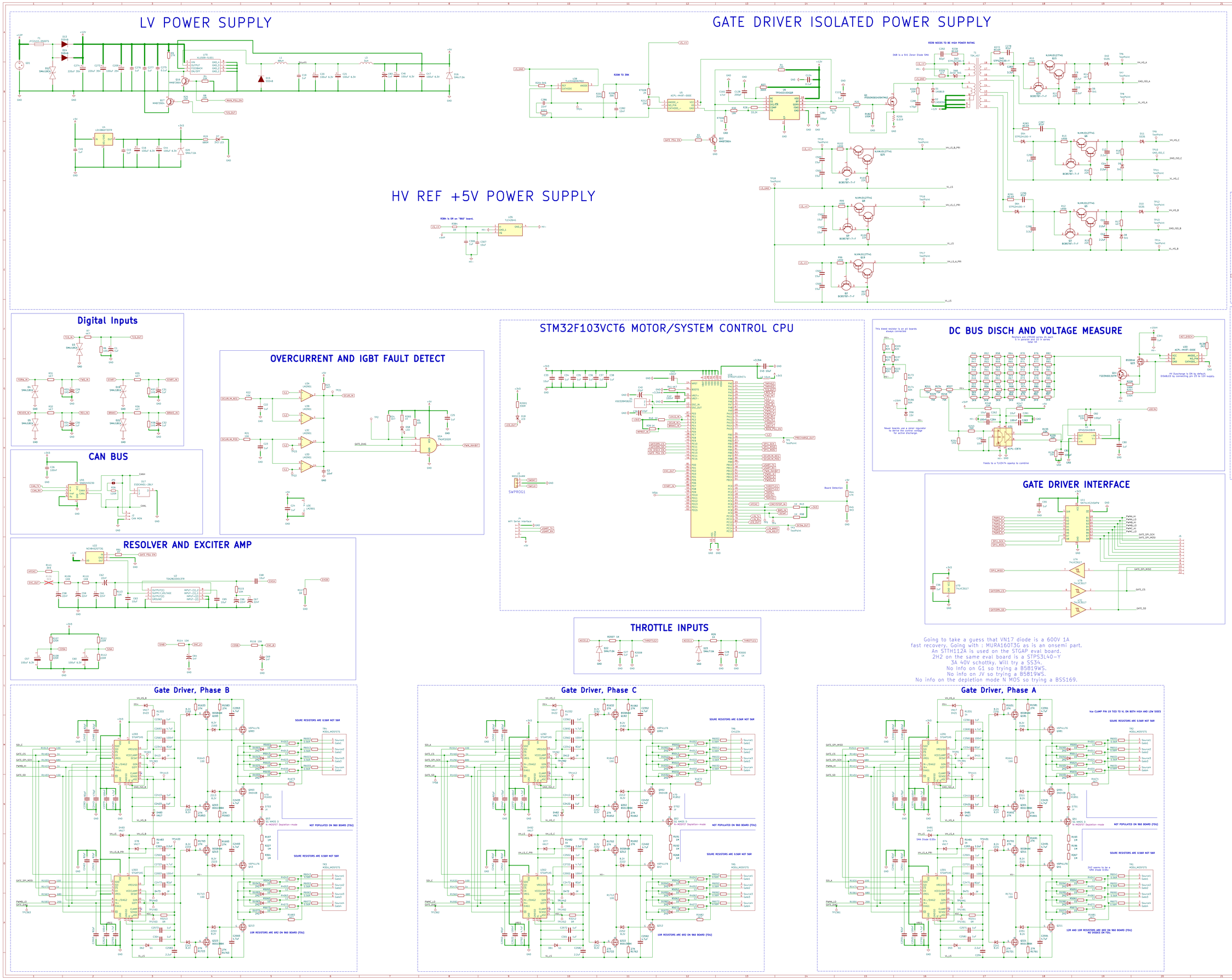
LV POWER SUPPLY
GATE DRIVER ISOLATED POWER SUPPLY
HV REF +5V POWER SUPPLY
HS TEMP SENSORS
approx 23k at 20c
drops on increasing temp
MOTOR TEMP SENSOR
Approx 150k at 20c.Drops on increasing temp.
Change R131,R132 to 1k.
Rx2 to 47K
OIL PUMP LINBUS
LINBUS ON USART4
CURRENT SENSORS
Current sensors are 73mV/mV variant
Digital Inputs
OVERCURRENT AND IGBT FAULT DETECT
STM32F103VCT6 MOTOR/SYSTEM CONTROL CPU
DC BUS DISCH AND VOLTAGE MEASURE
This bleed resistor is on all boards always connected
Feeds to a TLV247k opamp to combine
Newer boards use a zener regulator instead of a resistor for active discharge.
HV Discharge is ON by default.
DISABLED by connecting pin 23 to 12V supply.
CAN BUS
GATE DRIVER INTERFACE
CONNECTORS
COMPATIBLE WITH OEM HARNESS FOR RESOLVER AND ACCEL
RESOLVER AND EXCITER AMP
THROTTLE INPUTS
MOUNTING HOLES
Going to take a guess that VN17 diode is a 600V 1A
fast recovery. Going with : MURA160T3G as is an onsemi part.
An STTH112A is used on the STGAP eval board.
2H2 on the same eval board is a STPS3L40-Y
3A 40V schottky. Will try a SS34.
No info on G1 so trying a B5819WS.
No info on JV so trying a B5819WS.
No info on the depletion mode N MOS so trying a BSS169.
Gate Driver, Phase B
Gate Driver, Phase C
Gate Driver, Phase A
D.Maguire
A.Fishback
EVBMW.COM_OPENINVERTER.ORG
Project :
M3DU_BoardV3_2_kicad.sch
Title: M3DU Logic Board
Size: A3 | Date: 2025-08-27 | Rev: V1
KiCad 7.0.6 9.0.7 | Id: 1/1